

Design Verification Plan and Report - VBF-T5R1-DVPR-01

VBF-T5R1 delivery package - generated 2026-08-25

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Design Verification Plan and Report (virtual test version) - VBF-T5R1-DVPR-01

Case: t5_r1. Design under verification: R7B (`bridge/r7b\_negR261.json`), DFN precision, dual system Chen2020 + OKane2022.

Generation date: 2026-08-25. All result values mechanically from simulation outputs; determinations are `bda log-evaluate` round-7 verdicts (code-computed vs entry-0 criteria).

#	Verification item	Condition	Result value	Determination	Source
1	1C discharge capacity	1C (5 A nominal) discharge to 2.5 V, 25 ?C, DFN	6.2104 Ah (Chen2020) / 6.2086 Ah (OKane2022)	PASS (no capacity threshold in contract; nominal 5 Ah exceeded)	`cell/r7b_{chen,okane}_1c_dfn.json:capacity_ah`
2	Energy density	contract caliber (? layer mass, electrolyte excluded)	642.098 Wh/kg (Chen) / 641.118 Wh/kg (OKane)	PASS >= 500.94 Wh/kg	`cell/r7b_{chen,okane}_energy.json:energy_density_wh_kg`
3	4C fast-charge temperature rise	1C discharge to 2.5 V, then 20 A CC charge to 4.2 V, 45 ?C ambient, lumped thermal	T_max 326.829 K (Chen) / 328.189 K (OKane)	PASS <= 333.15 K (60 ?C)	`cell/r7b_{chen,okane}_4c_dfn.json:T_max_K`
4	4C fast-charge lithium plating	same protocol, plating enabled; criterion: anode surface potential < 0 V	anode min +0.01826 V (Chen) / +0.01798 V (OKane) -> plated = false	PASS (no plating), +18 mV margin	`cell/r7b_{chen,okane}_4c_dfn.json:anode_potential_v` (min>0 derived by log-evaluate)
5	Voltage window	parameter-set limits	2.5 ? 4.2 V (both sets)	PASS (contract definition)	parameter set dump
6	Nail penetration	-	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	-
7	Overcharge to thermal runaway	-	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	-
8	Crush / drop	-	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	-
9	Cycle life	-	N/A (aging protocol available on these sets but outside task contract; not executed - value not fabricated)	N/A	-
10	Rate-pulse internal resistance	-	N/A (only calc-energy DC-resistance caliber available: 3.50 m? Chen / 17.93 m? OKane, reported for information)	N/A	`calc-energy` dcr_ohm

Conclusion

All simulated verification items PASS under both parameter systems (Chen2020 and OKane2022). The three contract criteria - ED $\geq$ 500.94 Wh/kg, 4C no plating, $T_{\max} \leq 60$ °C - are each evidenced by file:key sources above and mechanically judged by `bda log-evaluate` (round 7, verdict=pass).

Uncovered items (paper limitations, cited directly): nail penetration, overcharge-to-thermal-runaway, crush, drop, cycle life, rate-pulse internal resistance - all N/A within the pure-simulation boundary of this case.

Known tool caveat: 4C-output `capacity\_ah` key under-reports charge capacity ~5% (library bug in pybamm_runner; `t?C\_rate/3600` missing the 5 Ah nominal factor). It is not used in any determination above (plating from `anode\_potential\_v`, $T_{\max}$ from lumped thermal, ED from 1C integration).